

Dice Roll Chance Experiment

I can perform repeated trials of a chance experiment and discuss the results. (ACMSP067)

Roll one dice 20 times and record each roll as a tally mark.

Equipment I will need:

- 1 x dice
- pencil
- activity sheet

Before you begin the experiment, make a prediction about what you think will happen. Think about what numbers may be rolled the most/least. Is there an even chance of rolling each number?

My prediction is: _____

Instructions:

1. Roll the dice
2. Record the number shown on the dice as a tally mark in the correct space in the table below.
3. Repeat steps 1 and 2 nineteen more times (so that you have rolled the dice 20 times).

Dice Roll Results:

Dice Number	Tally	Total
1		
2		
3		
4		
5		
6		

Was your prediction correct? Why/why not?

If you were to complete this chance experiment again do you think the results would be the same? Why/why not?

Complete the dice roll chance experiment again.

My revised prediction is: (What numbers do you think will be rolled the most/least, is there an even chance of rolling each number?)

Dice Roll Results:

Dice Number	Tally	Total
1		
2		
3		
4		
5		
6		

Dice Roll Questions

Was your revised prediction correct? Why/why not? (leave space/line for students to write their answer for each question)

Are the second tally results the same as your first dice rolls?

What is different?

What is the same?

Why do you think there is/is not a difference in the dice roll results?

If you were to complete this chance experiment again but roll the dice 50 times do you think the results would be similar? Why/why not?
